AgriRover-Q — Hardware Component List

Bill of materials, wiring and budgets for the Spot-style soil-survey quadruped. Generated from `agrirover/spec.py` and `config/hardware_pi.json`.

Size (walking)	303 × 217 × 282 mm	Mass (estimate)	1.95 kg
Legs	4 × (shoulder, hip, knee) = 12 servos	Belly clearance	140 mm walking
Battery	2S 7.4 V 5000 mAh LiPo	Runtime (estimate)	~62 min
Soil probe depth	50 mm (robot crouches)	Walking speed	~0.15 m/s

1. Bill of materials

Prices are rough hobby-retail estimates (USD) for budgeting only; printed parts are costed through the filament line.

#	Component	Qty	Function	Key specs	Interface	Mass (g)	Est. USD
COMPUTE & CONTROL							
1	**Raspberry Pi 4 Model B, 4 GB**	1	Main computer: runs the brain (gait, IK, navigation, perception, telemetry)	Quad A72 1.8 GHz, Wi-Fi, CSI camera port, 40-pin GPIO	USB-C 5 V	46	55
2	**microSD card, 32 GB A1**	1	OS + logs + offline upload spool	Raspberry Pi OS 64-bit	—	2	8
3	**Pi 4 heatsink + 30 mm fan**	1	Keeps the Pi from throttling during image processing		5 V GPIO	15	5
4	**PCA9685 16-ch PWM driver**	1	Generates 12 servo pulse trains (50 Hz)	12-bit, I2C 0x40	I2C	10	6
ACTUATION							
5	**DS3218 digital servo, 270°**	12	3 per leg: shoulder (abduction), hip, knee	20 kg·cm @ 6.8 V, 0.16 s/60°, metal gears, 25T spline	PWM via PCA9685	720	156
6	**25T aluminium servo horn (round)**	12	Joint couplers for thigh / shin / shoulder bracket		—	48	18
7	**625ZZ ball bearing**	12	Supports each joint on the side opposite the servo spline	5×16×5 mm	—	60	6
POWER							
8	**2S LiPo 7.4 V 5000 mAh, hard case**	1	Main battery (37 Wh)	≥ 30C, XT60, 138×46×25 mm	XT60	260	30
9	**Step-down converter 6 V / 20 A**	1	Servo rail (12 × DS3218 peak ~15 A)	In 7–9 V, out set to 6.0 V	screw terminals	40	12
10	**Step-down converter 5 V / 5 A, USB-C out**	1	Raspberry Pi supply	5.1 V, ≥ 5 A	USB-C	15	8
11	**XT60 connector pair**	1	Battery connection		—	4	2
12	**Rocker switch 20 A + 20 A blade fuse & holder**	1	Main power switch and short-circuit protection		—	10	5
13	**1000 µF / 16 V capacitor**	1	Bulk capacitance on the servo rail (stall spikes)	low ESR	—	2	0.50
14	**Resistors 10 kΩ + 3.3 kΩ**	1	Battery-voltage divider into ADS1115 A1 (8.4 V → 2.08 V)	1 %	—	—	0.20
SENSORS							
15	**BNO055 9-DoF IMU breakout**	1	Fused heading / roll / pitch: balance and navigation	I2C 0x28	I2C	3	35
16	**u-blox NEO-M8N GPS + patch antenna**	1	Route position, sample geotags	~2.5 m CEP, 5 Hz, NMEA	UART	25	20

#	Component	Qty	Function	Key specs	Interface	Mass (g)	Est. USD
17	**Raspberry Pi Camera Module 3**	1	Crop-row following + crop photos	12 MP, 66°×41° FOV, mounted 28° down	CSI	4	25
18	**Capacitive soil moisture sensor v1.2**	1	Soil volumetric water content (belly probe)	analog out, 3.3 V	ADS1115 A0	5	3
19	**DS18B20 waterproof temperature probe**	1	Soil temperature at probe depth	±0.5 °C, 12-bit, 1-Wire	GPIO4	10	3
20	**BME280 breakout**	1	Air temperature, humidity, pressure (in radiation shield)	I2C 0x76	I2C	2	8
21	**ADS1115 16-bit ADC**	1	Reads soil probe (A0) and battery voltage (A1)	I2C 0x48, gain 1 (±4.096 V)	I2C	3	8
22	**4.7 kΩ resistor**	1	1-Wire pull-up for DS18B20		—	—	0.10
STRUCTURE (3D PRINTED)							
23	**Body shell + lid + electronics deck**	1	220×100×70 mm box, orange top stripe	PETG, 3 mm walls, 40–45 % infill	—	230	—
24	**Shoulder bracket**	4	Holds the hip servo, driven by the shoulder servo	PETG	—	80	—
25	**Thigh (houses knee servo)**	4	100 mm link	PETG	—	140	—
26	**Shin**	4	110 mm link	PETG	—	100	—
27	**Foot pad**	4	Ø20 mm rubbery foot for grip on soil	TPU 95A	—	32	—
28	**Camera head, GPS mast, radiation shield, probe clamp**	1	Sensor mounts	PETG	—	37	—
29	**PETG filament 1 kg + TPU 250 g**	1	Material for all printed parts		—	—	30
WIRING & FASTENERS							
30	**Silicone wire 16 AWG (servo rail) + 22 AWG (signals)**	1	Power and signal wiring		—	60	8
31	**Servo extension leads 15 cm**	12	Servo to PCA9685		—	36	4.80
32	**M3 / M2.5 screw, nut, standoff kit**	1	Assembly; Pi on M2.5 standoffs		—	40	10
	Total					**2039**	**$467**

The list adds up to 2039 g, about 94 g above the 1945 g design estimate in section 4, because it budgets servo horns, bearings and wire more generously. Plan for ~2 kg.
Tools (not on the robot): **LiPo balance charger (2S)** (charging, storage mode); **LiPo safety bag** (charging and storage); **Multimeter** (set buck outputs before connecting loads); **3D printer** (PETG + TPU capable).

2. Wiring and interfaces

Raspberry Pi 4 40-pin header

Pin	Signal	Connects to	Notes
1	3V3	BNO055, BME280, ADS1115, soil probe, DS18B20 VCC	Sensors are all 3.3 V
3	GPIO2 / SDA1	I2C bus: PCA9685, BNO055, BME280, ADS1115	Set i2c_arm_baudrate=10000 for the BNO055 (clock stretching)
5	GPIO3 / SCL1	I2C bus	
7	GPIO4	DS18B20 data (1-Wire)	4.7 kΩ pull-up to 3V3; dtoverlay=w1-gpio
8	GPIO14 / TXD	GPS RX	enable_uart=1, disable serial console
10	GPIO15 / RXD	GPS TX	9600 baud NMEA
11	GPIO17	PCA9685 OE (optional)	Servo kill switch: high = outputs off
2 / 4	5V	GPS VCC (NEO-M8N module accepts 3.3–5 V)	Pi itself is fed via USB-C
6, 9, 14	GND	Common ground with servo rail and sensors	One star ground at the PCA9685
CSI	Camera	Pi Camera Module 3 (15-pin ribbon)	dtoverlay=imx708 (auto on Bookworm)

I2C devices and ADC channels

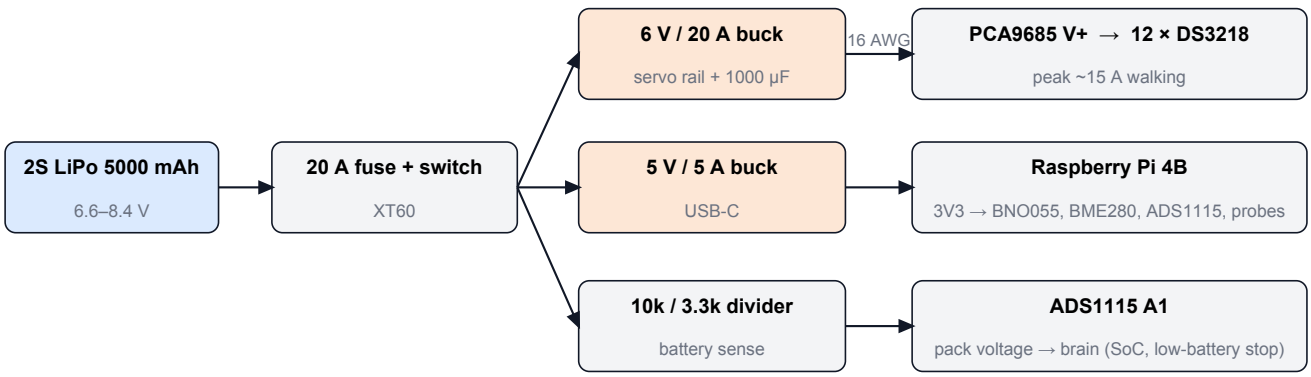
Device	Address	Role	Driver
PCA9685	0x40	12 servo channels (map below)	adafruit-circuitpython-servokit
BNO055	0x28	Orientation (NDOF fusion mode)	adafruit-circuitpython-bno055
BME280	0x76	Air T / RH / P	adafruit-circuitpython-bme280
ADS1115	0x48	A0 = soil probe, A1 = battery divider (×4.03)	adafruit-circuitpython-ads1x15

PCA9685 servo channel map

Ch	Joint	Dir	Zero (deg)	Ch	Joint	Dir	Zero (deg)
0	front left shoulder	+1	135	6	rear left shoulder	+1	135
1	front left hip	+1	135	7	rear left hip	+1	135
2	front left knee	+1	225	8	rear left knee	+1	225
3	front right shoulder	-1	135	9	rear right shoulder	-1	135
4	front right hip	-1	135	10	rear right hip	-1	135
5	front right knee	-1	225	11	rear right knee	-1	225

Zero angles are placeholders: calibrate each joint with the leg hanging straight down and write the result to config/hardware_pi.json.

Power distribution



3. Power budget

Load	Average power
12 servos: walking 35 W × 60%	21.0 W
12 servos: holding pose 8 W × 40%	3.2 W
Pi 4B + camera + WiFi (via buck)	3.9 W
sensors (GPS, IMU, BME280, ADC, probe)	0.4 W
Total (design estimate)	28.5 W
Usable energy: 5000 mAh × 7.4 V × 80%	29.6 Wh
Runtime estimate	**~62 min**

The simulator's servo model (mechanical power / 40 % efficiency + holding current) measured ~16 W average over a full survey, well under this 28.5 W design figure. Measure on the real robot before trusting either number; low-battery cut-off is set at 6.9 V under load.

4. Mass budget

Item	Mass
Servos ×12	720 g
Leg structure ×4	412 g
Battery (2S 5000 mAh)	260 g
Body shell (printed PETG)	230 g
Wiring + fasteners	100 g
Raspberry Pi 4B	61 g
6 V servo buck	40 g
GPS module	25 g
Soil probes + clamp	25 g
5 V Pi buck	15 g
GPS mast	15 g
BME280 + shield	12 g
PCA9685	10 g
Camera head mount	10 g
Camera module	4 g
BNO055 IMU	3 g
ADS1115 ADC	3 g
Total	**1945 g**

5. Bring-up checklist

• Set both buck converters with a multimeter **before** connecting the Pi or servos (6.0 V and 5.1 V).

• Enable I2C, UART and 1-Wire in raspi-config; check `i2cdetect -y 1` shows 0x28, 0x40, 0x48, 0x76.

• Calibrate servo zero angles one leg at a time with the robot on a stand (legs free).

• Calibrate the soil probe: record the ADS1115 reading in air and in water-saturated soil, enter them as `air_raw` / `water_raw` in the mission config.

• Walk a measured 5 m on the target soil and set `odometry_scale` = measured / commanded.

• Run the same mission JSON first in the simulator, then on the robot (`python -m agrirover.run_robot --mission …`).